QPI SC symmetry

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Contents

1 SC Basics **1**

2 QPI Basics **1**

3 Minimal model **2**

1 SC Basics

BdG Hamiltonian:

$$H(\boldsymbol{k}) = \begin{pmatrix} \varepsilon(\boldsymbol{k}) & \Delta(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & -\varepsilon(\boldsymbol{k}) \end{pmatrix} \tag{1}$$

$$= \begin{pmatrix} \frac{\Delta}{\sqrt{2E(E+\varepsilon)}} & \frac{\Delta}{\sqrt{2E(E-\varepsilon)}} \\ \frac{-(E+\varepsilon)}{\sqrt{2E(E+\varepsilon)}} & \frac{E-\varepsilon}{\sqrt{2E(E-\varepsilon)}} \end{pmatrix} \begin{pmatrix} -E & 0 \\ 0 & E \end{pmatrix} \begin{pmatrix} \frac{\Delta^*}{\sqrt{2E(E+\varepsilon)}} & \frac{-(E+\varepsilon)}{\sqrt{2E(E+\varepsilon)}} \\ \frac{\Delta^*}{\sqrt{2E(E-\varepsilon)}} & \frac{E-\varepsilon}{\sqrt{2E(E-\varepsilon)}} \end{pmatrix} \tag{2}$$

$$= U(\boldsymbol{k})\Lambda(\boldsymbol{k})U^\dagger(\boldsymbol{k}) \tag{3}$$

with $E = \sqrt{\varepsilon^2 + |\Delta|^2}$

Retarded Green's function:

$$G_0(\omega, \boldsymbol{k}) = \frac{1}{\omega + i\Gamma - H(\boldsymbol{k})} \tag{4}$$

$$= U(\boldsymbol{k}) \frac{1}{\omega + i\Gamma - \Lambda(\boldsymbol{k})} U^\dagger(\boldsymbol{k}) \tag{5}$$

2 QPI Basics

We consider a local impurity in the presence of the superconducting phase defined by the Hamiltonian

$$H = H_0 + V\,, \tag{6}$$

where H_0 is the unperturbed (impurity-free) superconducting Hamiltonian. V is the local impurity term, given as

$$V = \sum_{i\,\sigma\sigma'} [\hat{V}]_{\sigma\sigma'} \delta_{i,\,\boldsymbol{R}=0} c^\dagger_{i\sigma} c_{i\sigma'}\,, \tag{7}$$

where $\hat{V} = V_0\,\hat{I}$ for a scalar impurity and $\hat{V} = V_0\,\hat{\sigma}_z$ for a magnetic impurityThe QPI spectra are calculated using the T-matrix formalism

$$\delta\rho(\omega, \boldsymbol{r}) = -\frac{1}{\pi} \mathrm{Im}[\mathrm{Tr}[\hat{G}_0(\omega, \boldsymbol{r})\,\hat{T}(\omega)\,\hat{G}_0(\omega, -\boldsymbol{r})]]\,, \tag{8}$$

where the T-matrix can be shown to take the form:

$$\hat{T}(\omega) = \frac{\hat{V}}{1 - \hat{G}_0(\omega, \boldsymbol{r} = 0)\hat{V}}\,. \tag{9}$$

While this definition of the T-matrix is a formal definition, in practice it can be calculated as the non-trivial solution to the matrix equation

$$(1 - \hat{G}_0(\omega, \boldsymbol{r} = 0)\hat{V})\,\hat{T}(\omega) = \hat{V}\ . \tag{10}$$

Note that the Born approximation $\hat{T} = \hat{V}$ corresponds to a first-order approximation of the binomial expansion of the T-matrix. Subgap states (that exist in the superconducting gap) are associated with the poles of the T-matrix (at zero temperature)The condition for the existence of these bound states is

$$\det[1 - \hat{G}_0(\omega, \boldsymbol{r} = 0)\,\hat{V}] = 0\,. \tag{11}$$

Assuming a given unperturbed Green's function (Hamiltonian) and ω, this fixes the value of V_0 in Eqn. (7). Note that in principle this could introduce large V_0 values, so accuracy could potentially be improved by including bands higher in energy now accessible to the impurity states.

3 Minimal model

We consider a triangular lattice, with hopping t between nearest neighbour sites, and an onsite chemical potential μ. We add a variety of nearest neighbour superconducting pairings, corresponding to the different irreps of C_{6v} (which can be represented by Cooper pairs on the nearest neighbour bonds). For the moment, we will only consider even parity pairings, *i.e.*, the A_1, A_2, E_2 irreps. In reciprocal space, the Hamiltonian is given associated

$$H = \sum_{\boldsymbol{k},\sigma\sigma'} (\varepsilon(\boldsymbol{k}) - \mu)\delta_{\sigma\sigma'} c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma'} + \Delta(\boldsymbol{k})(i\sigma_y)_{\sigma\sigma'} c^\dagger_{-\boldsymbol{k}\sigma} c^\dagger_{\boldsymbol{k}\sigma'} + \Delta^*(\boldsymbol{k})(-i\sigma_y)_{\sigma\sigma'} c_{-\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma'} \tag{12}$$

$$\varepsilon(\boldsymbol{k}) = 2t \sum_{\alpha=1}^{3} \cos(\boldsymbol{a}_\alpha \cdot \boldsymbol{k}) \tag{13}$$

$$\boldsymbol{a}_1 = (1, 0), \qquad \boldsymbol{a}_2 = (-1/2, \sqrt{3}/2) \qquad \boldsymbol{a}_3 = -\boldsymbol{a}_1 - \boldsymbol{a}_2 \tag{14}$$

The even parity superconducting pairings on nearest neighbours can be expressed as:

$$\Delta(\boldsymbol{k}) = 2 \sum_{\alpha=1}^{3} \Delta_\alpha \cos(\boldsymbol{a}_\alpha \cdot \boldsymbol{k}) \tag{15}$$

$$\Delta^{A_1} = (1, 1, 1) \qquad \text{(extended } s\text{)} \tag{16}$$

$$\Delta^{E_2,1} = (1, -1/2, -1/2) \qquad (d_{x^2-y^2}) \tag{17}$$

$$\Delta^{E_2,2} = (0, -\sqrt{3}/2, \sqrt{3}/2) \qquad (d_{xy}) \tag{18}$$

$$\Delta^{E_2,+} = (1, e^{-2\pi i/3}, e^{2\pi i/3}) \qquad (d + id) \tag{19}$$

$$\Delta^{E_2,-} = (1, e^{2\pi i/3}, e^{-2\pi i/3}) \qquad (d - id) \tag{20}$$